

SilentFail-Bench: Beyond-Score Detection of Critic Blind Spots in Critic-Selected LLM-Agent Trajectories

Research Proposal — AAAI workshop style

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Abstract

When LLM agents commit by best-of- n (or related) selection under an imperfect critic, selection concentrates Goodhart errors: high selecting-critic score, verified task failure. We call these events *critic blind spots*. **Primary unit of analysis:** one completed *trajectory* τ ; gold $g(\tau) \in \{0, 1\}$ is trajectory-level task success (e.g., SWE-bench tests), never a claim that every intermediate step was invalid. **Open cell:** among critic-selected trajectories, does audit-panel uncertainty u improve held-out discrimination of $g=0$ *beyond* the selecting-critic score c , under a *disjoint* selection-critic vs audit-panel design? We propose **SilentFail-Bench**: a reproducible offline protocol with a frozen logistic combining model, repository-level train/test splits, a 200-trajectory pilot that sets power and go/no-go, and a quality-controlled diversity analysis (association, not causal mediation by default). A null is first-class and safety-relevant.

1 Motivation and Main Claim

Why it matters. Agent failures compound mid-task [14, 15, 13]. Many scaffolds select among candidates using a critic score [1, 2, 4]. Under Goodhart and the winner’s curse in best-of- n [9, 10, 11, 12], optimizing proxy c enriches the *selected* set for critic mistakes. Repair needs a trigger [5]. Prior UQ either changes *selection* (pessimistic BoN) [7, 11, 6], filters unconditionally, or uses agent verbal confidence (weak early) [16]. None answer: given the trajectory the agent already committed to, does a separate audit signal u still help beyond c ?

Main claim (selection-bias / Goodhart). Conditional on critic selection, disagreement among a *disjoint* audit panel is associated with residual information about trajectory failure beyond c . Lower pairwise error correlation across quality-matched panels is *associated* with larger residual detection (mechanism analysis; not a causal mediation claim unless diversity is randomized under fixed quality). Falsifiable null: even diverse panels share blind spots under selection.

What we do not claim. We do not invent “silent failures” [13]; do not propose a new auditor [14]; do not claim first use of RM uncertainty [6, 7, 11]. Novelty is the **selection-conditional, beyond-score, held-out estimand on agent trajectories with disjoint audit panels + pilot-powered SilentFail-Bench**.

2 Notation (locked)

- τ : one completed agent trajectory (primary observation).
- sel: the selection event that committed τ (best-of- n over candidate trajectories, or top- K integer retention from a fixed pool; defined per experiment).
- c : score from the *selection critic* c_{sel} on the committed τ (higher c = predicted higher quality / success).
- u : disagreement statistic from an *audit panel* (higher u = more disagreement).
- $g(\tau) \in \{0, 1\}$: trajectory-level gold; $g=1$ iff verified task success (unit tests / env reward). Positive class for AUROC: $g=0$ (failure).
- n : candidate-pool size for best-of- n (select top-1).
- K : integer number retained under top- K filtering (not a percent of small n).
- m : audit-panel size.
- b : selective-verification budget (fraction of selected trajectories flagged for review).

3 Related Work: Crowded Neighborhood, Open Cell

Neighbor	What they do	Difference
False success [13]	Self-declared done vs env gold	Label = agent claim, not critic-selected high- c trajectory
AgentForesight; de- tect+repair [14, 15]	Online auditors / telemetry + rollback	Not held-out c vs $c+u$ under critic sel
Last Step Matters [16]	Early verbal/perplexity UQ weak	Different u ; not critic-panel residual test
CAUTION; mini-N [11, 12]	UQ changes BoN <i>selection</i> / guards false accept	Selection policy $\neq$ post-selection residual detection
URM/URME; ensembles; RM→judge [6, 7, 8, 17]	Unconditional UQ / filter / route prefs	Non-agentic or not conditioned on agent sel
PRMs / AgentPRM [1, 4, 3]	Score and select	No beyond-score held-out Δ AUROC on selected trajectories
Judge disagreement in RLHF [18]	Taxonomy of hacking / judge disagreement	Training dynamics, not agent-trajectory post-selection audit

Critical gap. To our knowledge, no public evaluation jointly reports: (i) held-out Δ AUROC of audit uncertainty beyond selecting-critic score, (ii) conditional on critic selection of LLM-agent trajectories, (iii) trajectory-verified gold, (iv) a disjoint selection-critic vs audit-panel design, and (v) quality-controlled diversity comparisons. Closest work uses uncertainty to *change* BoN selection [11, 6, 12], not to detect residual failure *after* selection.

4 Gold Label and Unit of Analysis

Primary endpoint. One observation = one selected trajectory τ . $g(\tau)$ is trajectory-level verified success/failure. A failed final test suite means the *trajectory* failed; it does **not** prove that every intermediate action was invalid. A *critic blind spot* is a selected trajectory with high c and $g=0$.

How many observations. Each trajectory contributes **exactly one** primary observation (c, u, g , controls). Train/test splits and bootstrap CIs are clustered by trajectory (and by repository when splitting).

Secondary (optional, clearly approximate). Step-level claims use Math-Shepherd-style Monte-Carlo rollouts from a state (“still on a succeeding path”) [2], reported with rollout-noise sensitivity and a hierarchical model clustered by trajectory. Step labels are never treated as exact oracles.

5 Selection Critic vs Audit Panel (Leakage Control)

Disjoint (primary). Selection critic c_{sel} chooses τ . A separate frozen audit panel $\{c_1, \dots, c_m\}$ with $c_{\text{sel}} \notin \{c_1, \dots, c_m\}$ produces u . This is the clean beyond-score test.

Shared (ablation only). c_{sel} is included in the panel (deployment-realistic, coupled). Reported separately; not the primary confirmatory contrast.

Definitions of c and u (primary).

- $c := c_{\text{sel}}(\tau)$, the selecting critic’s score on the committed trajectory.
- Rank-normalize each audit critic’s scores using *training-split* rank parameters only (frozen at test time).
- Primary $u :=$ sample standard deviation of the m rank-normalized audit scores on τ .
- Exploratory u : range; mean pairwise rank disagreement (labeled exploratory).

Combining model (frozen, named). Let $x_{\text{ctrl}} = (\text{length}, \text{step-count}, \text{mean agent entropy})$. Fit on training repositories only:

$$\begin{aligned} M_c : \quad \Pr(g=0) &= \sigma(\beta_0 + \beta_c c + \beta_{\text{ctrl}}^\top x_{\text{ctrl}}), \\ M_u : \quad \Pr(g=0) &= \sigma(\beta_0 + \beta_u u + \beta_{\text{ctrl}}^\top x_{\text{ctrl}}), \\ M_{c+u} : \quad \Pr(g=0) &= \sigma(\beta_0 + \beta_c c + \beta_u u + \beta_{\text{ctrl}}^\top x_{\text{ctrl}}). \end{aligned}$$

Held-out AUROCs use predicted failure probabilities.

$$\Delta\text{AUROC} := \text{AUROC}_{\text{held-out}}(M_{c+u}) - \text{AUROC}_{\text{held-out}}(M_c).$$

Also report M_u alone, AUPRC (class imbalance), LR test for β_u in M_{c+u} , and trajectory-/repo-clustered bootstrap 95% CIs for ΔAUROC .

Confirmatory vs exploratory. **Confirmatory:** disjoint-panel ΔAUROC at the anchor (diverse vs quality-matched seed panel). **Exploratory:** shared-panel ablation; alternate u stats; secondary step labels; full pressure×diversity grid; E4–E7. Multiple exploratory slices use BH-FDR or are clearly marked non-confirmatory.

6 Pilot, Power, and Compute

200-trajectory pilot (Stage 0; before full matrix). Anchor configuration only. Report: #trajectories; #selection decisions logged; selected-set failure count and rate with CI; c and u distributions; c – u correlation (range restriction); exclusions.

Go/no-go (numeric thresholds set from pilot, not hopes). Continue to Stage 1 only if: (i) enough $g=0$ cases in the selected set for useful precision on ΔAUROC (pilot simulation sets the count), and (ii) M_c alone does not already saturate (held-out AUROC headroom for u). Planning assumption only (not a result): failure rates on hard agent benches are often non-trivial; the pilot replaces any 10–20% target.

Power. Simulation-based power using observed pilot base rate, within-repo clustering, and the train/test split. Predeclare minimum detectable ΔAUROC worth pursuing (set after pilot; planning placeholder $\Delta\text{AUROC} \geq 0.03$ pending pilot). Report required trajectories/failures for 80% and 90% power.

Stage	Rollouts	Critic calls (order)	Shrink-if-needed
Pilot (200 traj.)	$200 \times n=16$ gens	$200 \cdot 16$ select + $200 \cdot m$ audit	Cut n or m first
Anchor study	Pilot-informed N_τ	$\propto N_\tau \cdot (n + m)$	Drop WebArena; $m=2$
Full matrix	Anchor $\times$ sweeps	$\times$ diversity $\times$ n -grid	Exploratory only; FDR

Compute budget (planning; refine after pilot). Report tokens/GPU-hours, wall time, and \$ cost after the pilot’s measured per-trajectory cost. If pilot cost or base rate makes the matrix infeasible, keep confirmatory anchor only.

7 Selection-Pressure Specification

Best-of- n and top- K are **not** interchangeable.

Best-of- n (primary pressure axis). Candidate pool size = n ; retain exactly 1 (argmax of c_{sel}). Sweep $n \in \{4, 8, 16, 32\}$. Anchor: $n=16$ (not paired with a percent).

Top- K filtering (separate experiment). Fix pool size $M=64$ candidates; retain integer $K \in \{1, 2, 4, 8\}$. Never write “ $k=5\%$ of $n=16$ ” (undefined / sub-integer).

For every pressure cell, the paper will list (M or n , retained, rule).

8 Diversity vs Critic Quality

Diversity interventions (seed / data / base / public-zoo) are separate conditions, not one assumed ordinal scale unless the pilot supports ordering.

Before comparing panels on ΔAUROC :

- Match or regress on per-critic quality (RewardBench-style accuracy/calibration and/or held-out ranking accuracy on train repos).
- Report per-critic accuracy alongside pairwise error correlation on gold.
- Primary diversity contrast: quality-matched seed panel vs quality-matched base-diverse panel at fixed m .

Default claim language: error correlation is *associated* with residual detection. “Mediation” reserved only if treatment (diversity) is randomized under fixed quality with stated assumptions; otherwise “mechanism analysis.”

9 Preregistered Anchor

- **Tasks:** SWE-bench Verified; split by repository (70/30, seed 20260922).
- **Agent scaffold:** mini-swe-agent (commit pinned in `silentfail_bench/configs/anchor.yaml` before pilot).
- **Score rule:** c = selection-critic scalar on the *full* committed trajectory text (messages + patch); BoN ranks full trajectories; no per-step mean.
- **Selection critic:** Skywork-Reward-V2-Qwen3-8B (HF id in anchor config); $c = c_{\text{sel}}(\tau)$.
- **Audit panel (disjoint):** $m=4$ public RMs listed in anchor config; none equal to c_{sel} .
- **Selection rule:** best-of- n with $n=16$, retain top-1; $T=0.7$; max steps 50; timeout 1800s.
- u : std. of train-fit rank-normalized audit scores.
- **Combining model:** logistic M_c , M_{c+u} ; controls = length, step-count, mean entropy.
- **Compute:** vLLM OpenAI-compatible server and/or hosted API; logistic fit on CPU.
- **Primary endpoint:** held-out ΔAUROC (failure positive).
- **Primary contrast:** quality-matched diverse vs seed audit panel (disjoint).
- **CI:** repository-clustered bootstrap 95%.
- **Exclusions:** empty candidate pools; critic NaN; timeout; max steps exceeded.
- **Stage-0 code:** `silentfail_bench/` (schema, mock/vLLM runner, pilot report, power helpers).

10 Experiments

ID	Maps to / design / metric
E0 Pilot	Vacuousness + power inputs: failure rate, c/u dist., $c-u$ corr., M_c saturation; go/no-go
E1	Confirmatory beyond-score: held-out ΔAUROC at anchor; exploratory vs $n \in \{4, 8, 16, 32\}$
E2	Diversity vs seed at fixed quality/ m ; association with error correlation (mechanism)
E3	Pressure $\times$ diversity: ΔAUROC vs $\log n$ (operational n); exploratory
E4	Cheap u (dropout / quantile / self-consistency) vs ensemble; exploratory
E5	Trajectory failure taxonomy labels; per-class ΔAUROC ; exploratory
E6	WebArena / ALFWorld generality; exploratory if compute allows
E7	Offline selective verification (not adaptive recovery): fixed budget $b \in \{5\%, 10\%, 20\%\}$; rank by M_{c+u} vs M_c vs M_u vs length/entropy/step-count; report failures caught, false alarms, precision@ b , recall@ b , cost per extra failure caught; alert action = queue for human review / alternate critic / stop (simulated)

Controls vs baselines. Length, step-count, and entropy enter M_c/M_{c+u} as covariates so residual u is not a proxy for verbosity or horizon. The same features also appear as operational baselines in E7.

Robustness. Repo holdouts; AUPRC; calibration (ECE); sensitivity to u statistic, normalization, panel composition; report selected-set range restriction ($c-u$ corr.) as a result.

11 Ideal Results (pilot-informed)

1. Pilot yields a measurable selected-set failure rate with CI (replaces any fixed 10–20% hope).
2. Disjoint-panel held-out $\Delta\text{AUROC} > 0$ at anchor with clustered CI excluding 0 (or a clean null).
3. Quality-matched lower error correlation is associated with larger ΔAUROC .
4. Exploratory: flatter ΔAUROC decay in $\log n$ for diverse vs seed panels.
5. E7: at fixed b , M_{c+u} catches more failures per review than M_c alone.
6. Alt. ideal: diverse-panel null $\Rightarrow$ negative-result safety paper [8].

12 Limitations and Staging

Trajectory g does not credit-assign steps; secondary step labels are noisy. Offline E7 $\neq$ adaptive recovery. Critic scale mismatch mitigated by train-only rank-norm, not eliminated. Logging cost may force confirmatory-only scope. Range restriction may erase headroom (informative). Scaffold/critic hashes must be frozen before Stage 0.

Stage 0: 200-traj. pilot + go/no-go + power + cost. **Stage 1:** confirmatory anchor only. **Stage 2:** exploratory matrix as affordable.

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